

Polynomial

*Declare: All polynomials here are real polynomials with real coefficients!!!

Polynomial Functions of x : a family of functions, each of which can be written as a sum of non-negative integer powers of x . General form:

$$\sum_{i=0}^n a_i x^i. \text{ terms: } a_i x^i \text{ coefficients: } a_i \text{ leading term: } a_n x^n \text{ leading coefficient: } a_n \text{ degree: } n$$

Zero, Root, and Factor: If $f(a) = 0$, then $x = a$ is a zero of $y = f(x)$ or a root of $f(x) = 0$, and $(x - a)$ is a factor of $f(x)$ in which $f(x) = (x - x_1)(x - x_2)\dots(x - a)\dots(x - x_n)$.

Nested form: $\sum_{i=0}^n a_i x^i = f(x) = x(x(\dots(x(a_n x + a_{n-1}) + a_{n-2})\dots) + a_1) + a_0$. (Nested by $n - 1$ times). **Synthetic substitution** is the process of calculating the value of $f(x)$ by using the property of nested form.

Remainder Theorem: A polynomial function with degree higher than 0/ n can be written as: $P(x) = (x - a)Q_0(x) + r/\prod_{i=0}^n (x - a_i)Q_n(x) + \sum_{i=0}^n r_i x^i$; Which the reminder function is: $R_n(x) = \sum_{i=0}^n r_i x^i$. For divisor $x - a$, the remainder is $P(a) = r = R(a)$.

Synthetic Division: To divide $P(x) = \sum_{i=0}^n a_i x^i$ by $x - a$. We start with writing down all the coefficients in decreasing degree order on a row. We first write down the leading coefficient on the lowest column, times it by a and write it down below the next coefficient, and add them up. Again, we write it on the lowest column. Then, we repeat the step of adding and timing. At last we have $n + 1$ numbers, where the last number is the remainder. The first n numbers are the coefficients for its quotient polynomial in decreasing degree order.

Factor Theorem: If $r = R(a) = P(a) = 0$, then $x - a$ is a factor of $P(x)$.

Location Principle: If $f(x)$ is continuous on $[a, b]$ and $f(a)f(b) < 0$, then $\exists c \in (a, b)$ such that c is a root.

Factorization, Rational Root Theorem: For $\sum_{i=0}^n a_i x^i = 0$ and $a_k \in \mathbb{Z}$, $x = \frac{p}{q}$. Then p is a factor of a_0 and q is a factor of a_n . $(qx - p)$ is a factor of $\sum_{i=0}^n a_i x^i = 0$.

Sum and product root theorem: $\sum_{i=0}^n a_i x^i = 0 = a_n \prod_{i=1}^n (x - x_i)$. $\{x_n\}$ are the roots of the equation. Then,

$$\sum_{1 \leq i_1 < \dots < i_k \leq n \ (i_n \in \mathbb{Z})} (\prod_{j=1}^k x_{i_j}) = (-1)^k \frac{a_{n-k}}{a_n} \ \forall k \in \{x | x \in \mathbb{Z}^+, x \leq n\}$$

Fundamental Theorem of Algebra: Every non-zero, single-variable, degree n polynomial with complex coefficients, $P(x) = \sum_{i=0}^n a_i x^i$, has, counted with multiplicity, exactly n roots over $\mathbb{C}$.

Conjugate root Theorem: If $a + \sqrt{b}$ is a root of $P(x) = 0$ for $a_k \in \mathbb{Q}$, then $a - \sqrt{b}$ is also a root of $P(x) = 0$. If $a + bi$ is a root of $P(x) = 0$ for $a_k \in \mathbb{R}$, then $a - bi$ is also a root of $P(x) = 0$.

Quadratic Formula: $ax^2 + bx + c = 0$. $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. $\Delta = b^2 - 4ac$. If $\Delta > 0$, then there are two real roots. If $\Delta = 0$, then there is a pair of equal real roots (double root). If $\Delta < 0$, then there are two imaginary roots.

Inequality

Fundamental Theorem of Inequality: $a > b$, iff $-a < -b$.

Polynomial Inequalities: solve the polynomial equation, factorize the polynomial, and check the sign. A sign graph may help you analyze.

Absolute Value Inequalities: Consider the geometric meaning of $|x - a|$, the distance of x to a . Use the property that $|a| > 0$, $|ab| = |a||b|$, and $|a + b| \leq |a| + |b|$. Discuss circumstances of $a > 0$, $a = 0$, and $a < 0$, or partition the real-axis.

Fractional Inequalities: Remember that the denominator must not be 0, but solve the roots of it. Analyze its sign just like a polynomial.

Radical Inequalities: Be aware that $\sqrt{f(x)}$ exist only when $f(x) \geq 0$. Isolate one square root and square both sides until getting rid of all square roots.

Exponential, Logarithmic, Trigonometric or other functional Inequalities: a^x and $\log_a x$ are increasing functions for $a > 1$, decreasing for $0 < a < 1$. Use exponential or logarithmic calculations to simplify the inequality and directly take the logarithm or exponent of the two sides with equal base greater than 1. For trigonometry, drawing a graph may help. Sometimes substituting may generate a new polynomial inequality, and you may solve its range.

Transformations of Graphs

Translations: The graph $y = f(x) + c$ is the graph of $y = f(x)$ translated up by c units. If c is negative the graph is translated down. The graph $y = f(x + d)$ is the graph of $y = f(x)$ translated left by d units. If d is negative the graph is translated right.

Stretches: The graph $y = pf(x)$ is the graph of $y = f(x)$ stretched vertically relative to the x -axis with scale factor p . When $p > 1$, this is a stretch away from the x -axis. If $0 < p < 1$, then $y = f(x)$ is compressed vertically relative to the x -axis (towards). If $p < 0$, the scale factor is negative ($-p$), and it might be easier to think of the transformation as a stretch/compression by scale factor p followed by reflection in the x -axis. The graph $y = f(qx)$ is the graph of $y = f(x)$ stretched horizontally relative to the y -axis scale factor $\frac{1}{q}$. This can be considered a compression relative to the y -axis (towards) when $q > 1$. When $0 < q < 1$, it is considered a stretch relative to the y -axis (away from), and when $q < 0$, the scale factor is negative, and it is easier to think of the transformation as a stretch/compression by scale factor followed by a reflection in the y -axis.

Trigonometric Functions

*Be aware of the mode of your calculator!!!

Radian and Degree: A full turn is 2π radian or 360° . For a unit circle, the measure of the angle in radians of a sector is equal to the arc length, yet it is defined as the ratio of the arc length to the radius of the circle, and it is dimensionless.

Sine and Cosine: Starting with a real number α , first mark a point A on the unit circle representing the number α . The sine of the number α , written $\sin \alpha$, is the distance of the point A from the horizontal axis. The cosine of the number α , written $\cos \alpha$, is the distance of the point A from the vertical axis.

Period and Amplitude: If $f(x) = f(x + p) \forall x \in \mathbb{R}$, then $f(x)$ is a **periodic function** with period p . Its amplitude is $(\max\{f(x)\} - \min\{f(x)\})/2$. The sine and cosine functions are periodic with period 2π and have amplitude 1.

Trigonometric Identities ($k \in \mathbb{Z}$):

$$\begin{aligned} \sin x &= \sin(\pi - x) = \sin(x + 2\pi) & \sin(x + \pi) &= \sin(-x) = -\sin x \\ \cos x &= \cos(-x) = \cos(x + 2\pi) & \cos(\pi - x) &= \cos(\pi + x) = -\cos x \\ \cos\left(\frac{\pi}{2} - x\right) &= \sin x & \cos\left(\frac{\pi}{2} + x\right) &= -\sin x \\ \sin\left(\frac{\pi}{2} - x\right) &= \cos x & \sin\left(\frac{\pi}{2} + x\right) &= \cos x \\ \sin x &= \sin(x + 2k\pi) = \sin(\pi - x + 2k\pi) & \cos x &= \cos(x + 2k\pi) = \cos(-x + 2k\pi) \end{aligned}$$

Tangent: The ratio between the sine and cosine function. Have period π and no amplitude.
 $\tan x = \tan(x + k\pi)$

Special values:

Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Inverse trigonometric Functions: The inverse function of $f(x) = \sin x$ is $f^{-1}(x) = \arcsin x$ or $\sin^{-1} x$. Its domain is $[-1, 1]$ and its range $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The inverse function of $f(x) = \cos x$ is $f^{-1}(x) = \arccos x$ or $\cos^{-1} x$. Its domain is $[-1, 1]$ and its range $[0, \pi]$. The inverse function of $f(x) = \tan x$ is $f^{-1}(x) = \arctan x$ or $\tan^{-1} x$. Its domain is $\mathbb{R}$ and its range $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Law of Sine: In triangle $\triangle ABC$, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2r$. r is the radius of the circumscribed circle of the triangle. Area formula: $\frac{1}{2}ab \sin C$

Law of Cosine: In triangle $\triangle ABC$, $c^2 = a^2 + b^2 - 2ab \cos C$.

Compound Angle Identities:

$$\begin{aligned} \sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y & \cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y \\ \tan(x \pm y) &= \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y} \end{aligned}$$

Functions of the form $a \sin x + b \cos x$: $a \sin x + b \cos x = R \sin(x + \alpha) = R \cos(x + \beta)$, in which $R^2 = a^2 + b^2$, $R \cos \alpha = -R \sin \beta = a$, and $R \sin \alpha = R \cos \beta = b$.

Reciprocal Trigonometric Functions: secant: $\sec x = \frac{1}{\cos x}$, cosecant: $\csc x = \frac{1}{\sin x}$, and cotangent: $\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$.

Pythagorean Identity: $\sin^2 x + \cos^2 x = 1$; $\sec^2 x = 1 + \tan^2 x$; $\csc^2 x = 1 + \cot^2 x$.

Trigonometric Identities:

$$\sin a + \sin b = 2 \sin \frac{a+b}{2} \cos \frac{a-b}{2}$$

$$\sin a - \sin b = 2 \cos \frac{a+b}{2} \sin \frac{a-b}{2}$$

$$\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2}$$

$$\cos a - \cos b = -2 \sin \frac{a+b}{2} \sin \frac{a-b}{2}$$

$$\sin^2 a = \frac{1 - \cos 2a}{2}$$

$$\cos^2 a = \frac{1 + \cos 2a}{2}$$

$$\tan^2 a = \frac{1 - \cos 2a}{1 + \cos 2a}$$